Homework Flow — Cost Verification & Provider Reconciliation

Purpose. This document lets us *confirm* the per-session cost the estimator shows for the Homework Help (Tier 2) flow. It checks every provider unit price against the live provider documentation, then shows exactly how the biggest line — **TTS audio** — is now counted on the **measured average** narration length, how image generation is costed at **one diagram per session**, and explains the o3 reasoning-token risk.

Bottom line up front: the unit prices are *correct and current*. Audio is now billed on the **MEASURED AVERAGE** narration length (Section 2), route-specific — measured across real homework sessions rather than the old conservative 24,000-char maximum. Every price carries a **+15% buffer** and Inworld is priced on the **Developer plan ($17 / 1M characters)**. Image generation is a single diagram per session (Section 3), a minor line. The figure to keep validating on a real invoice is o3's **hidden reasoning tokens** (Section 4.3).

1. Provider unit prices — confirmed against documentation (July 2026)

Service	Model	Published list price	In estimator	Confirmed
Reasoning LLM	**o3**	**$2.00 / 1M input · $8.00 / 1M output**	$2 / $8	✅ matches OpenAI
Prose / vision LLM	**gpt-4.1**	$2.00 / 1M input · $8.00 / 1M output	$2 / $8	✅ matches OpenAI
Diagram image	**Gemini 3.1 Flash Image**	$0.067 / image (standard-tier list)	**$0.10** (conservative round-up)	✅ ≥ list
Text-to-speech	**Inworld TTS-2**	**$17.00 / 1M characters (Developer plan)**	$17	✅ matches Inworld
Safety	omni-moderation	Free	$0	✅

The +15% safety buffer. Every price above is multiplied by **1.15** inside the estimator before any figure is shown, as an internal margin for retries, prompt drift and future price moves. So the effective rates the estimator charges against are: o3 / gpt-4.1 **$2.30 / $9.20** per 1M, Gemini **$0.115**

per image, Inworld **$19.55** per 1M characters. Throughout, we show **both** columns — "*list*" (the provider's published price, for reconciliation) and "*+15%*" (what the estimator displays).

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2. Text-to-speech (Inworld TTS-2) — billed on the MEASURED AVERAGE narration length

Inworld bills **per character** sent to synthesis. Earlier this document costed audio on a conservative **24,000-char maximum**. We have since **measured** the characters the homework flow actually sends to TTS across real sessions (`homework_tts_char_count.py`, 6 sessions), and the estimator now costs audio on that **measured average** — and, crucially, **route-specific**: prose *explanation* sessions narrate 2–4× longer than *calculation* sessions, so the blended figure follows the calculation / explanation mix.

Route (what the question needs)	Measured avg TTS chars / session	% of the 24,000 ceiling	n
Calculation → o3 (Maths / Physics / Chemistry sums)	**4,758**	20%	3
Explanation → gpt-4.1 (English, Biology prose)	**12,611**	53%	3
Blended average (50/50 default)	**8,685**	36%	6

Cost of the blended 8,685-char average: list `8,685 ÷ 1,000,000 × $17` = **$0.1476** · +15% = **$0.1698**. Per route: calculation `4,758 × $17/1M` = **$0.0809** list / **$0.0930** (+15%); explanation `12,611 × $17/1M` = **$0.2144** list / **$0.2465** (+15%).

The 24,000-char maximum remains a safe worst-case bound, but the measured average is ~**2.8× lower**, so it is the realistic figure for a blended forecast. Inworld also gets cheaper with commitment, so a real audio invoice can only come in lower than the Developer-plan rate:

Inworld TTS-2 tier	$ / 1M characters	Cost of the 8,685-char avg session
On-demand	$25.00	$0.217
Developer (what we estimate on)	**$17.00**	**$0.148**
Higher commit	$12.50	$0.109
Enterprise	$10.00 (as low as $5)	$0.087 (↓ $0.043)

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3. Image generation (Gemini 3.1 Flash Image) — one diagram per session

Homework Help generates **exactly one diagram image per session**, regardless of walkthrough depth or how the question arrived. This is homework-only behaviour: a single supporting diagram is produced for the session and reused across the walkthrough steps rather than re-generated at each phase.

```
images / session = 1   (fixed — one diagram per session)
```

Because it no longer scales with walkthrough depth, this line is flat and small. The estimator prices the image at **$0.10** — a conservative round-up from Gemini's ~$0.067 standard-tier list, for headroom:

Images / session	Cost (list @ $0.10)	Cost (+15%)
1 (fixed)	**$0.10**	**$0.115**

> **This is a minor provider line in a session.** At one image, generation is **$0.10 list / $0.115 buffered**. If in future the flow needs more than one diagram per session, raise the image assumption in the estimator and this line scales linearly.

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4. o3 reasoning model — tokens and cost

o3 is only used for **calculation / derivation** questions (Maths, Physics, Chemistry, computational Science). Prose/explanation questions run on gpt-4.1. The build pipeline is five LLM calls (steps A · parallel B · demo C · 3A explanation · answer key D); A→B→C can regenerate on a self-check failure, so they carry a retry factor (default **1.15×**).

4.1 Per-call token assumptions (o3 route)

Output tokens below are set to **include** o3's hidden reasoning tokens — see Section 4.3.

Call	Input tok	o3 output tok	Runs / session	o3 cost (list)
A · Solution steps	700	3,000	1.15	$0.0292
B · Parallel question	800	2,500	1.15	$0.0248
C · Parallel demo (worked)	900	3,000	1.15	$0.0297
3A · Explanation	600	2,200	1.00	$0.0188
D · Private answer key	900	3,000	1.00	$0.0258
Validate attempt	700	1,500	3.00	$0.0402
Follow-up Q&A	800	1,500	1.00	$0.0136
o3 subtotal	**~7,160**	**~20,975**		**$0.1821**

Plus the fixed gpt-4.1 intake/classify calls that run in **every** session (image extract + analyze + match): **$0.0118**.

4.2 Per-session o3 cost (the math)

```
o3 input :  7,160 tok ÷ 1,000,000 × $2  = $0.0143
o3 output: 20,975 tok ÷ 1,000,000 × $8  = $0.1678
o3 total (list)                          = $0.1821  →  +15% = $0.2094
```

4.3 Why o3 can look cheaper than the docs — hidden reasoning tokens

The unit price is right ($2/$8); the risk is the token count, not the rate. o3 bills *invisible* reasoning tokens at the **output** rate. OpenAI's documentation notes a single o3 answer can burn **5,000–20,000 internal reasoning tokens**. Our model already assumes ~21,000 o3 output tokens for a whole calculation session (near the high end), but if o3 reasons *harder*, cost scales. This table shows a full **calculation** session (o3 LLM + the measured-average calc audio $0.081 + the one-image $0.10 line) as reasoning load rises:

o3 reasoning load	o3 output tok / session	Calc session (list)	Calc session (+15%)
As modelled	~21,000	$0.375	**$0.431**
1.5× harder	~31,500	$0.459	$0.528
2× harder	~42,000	$0.543	$0.624
3× harder	~63,000	$0.711	$0.818

> The o3 output-token counts are **editable** in the estimator's *Advanced assumptions* — tune them to a sample of real billed sessions and the estimate tracks reality.

5. Full homework-session confirmation table

Per-session cost by service, with **audio at the measured average** (route-specific) and **one real-time image**. **Calculation** sessions use o3; **explanation** sessions use gpt-4.1; the **blended** row is the estimator default (50/50, image intake, audio on).

Session type	o3	gpt-4.1	Gemini image (×1)	Inworld audio (avg)	Total (list)	Total (+15%)
Calculation (o3)	$0.182	$0.012	$0.100	$0.081	**$0.375**	**$0.431**
Explanation (gpt-4.1)	—	$0.070	$0.100	$0.214	**$0.384**	**$0.442**
Blended (default)	$0.091	$0.041	$0.100	$0.148	**$0.380**	**$0.437**

Every number is reproducible from Sections 2–4 with the published unit prices — that is the confirmation: **provider price × quantity = the estimator's figure.** Audio is now a measured line (~35–55% of a session depending on route), not a worst-case ceiling; the single image line is minor.

6. Scaling to volume (monthly / yearly)

Using the **blended buffered** cost of **$0.437** per session, scaled by each preset's monthly session count (the *range* uses the calculation-only and explanation-only bounds, $0.431 → $0.442):

Preset	Students × questions	Sessions / month	Monthly (blended)	Yearly (blended)	Monthly range
Reset to defaults	200 × 10	2,000	**$874**	**$10,488**	$862 – $884
Light usage	300 × 20	6,000	**$2,622**	**$31,464**	$2,586 – $2,652
Heavy usage	600 × 40	24,000	**$10,488**	**$125,856**	$10,344 – $10,608

Monthly figures use the blended-default per-session cost. The two levers that move these most: the explanation / calculation mix (prose sessions narrate 2–4× more audio) and the o3 reasoning-token count (Section 4.3). The Inworld commitment tier lowers audio further ($17 Developer → $12.50 higher commit → $10 enterprise).

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7. Reconciliation summary

1. **Unit prices are correct and current** (verified July 2026) — o3 & gpt-4.1 at $2/$8 per 1M tokens, Gemini image at $0.067, Inworld TTS-2 at **$17 per 1M characters (Developer plan)**.
2. **Audio is now a measured-average line, route-specific** — calculation ~4,758 chars, explanation ~12,611 chars, blended ~8,685 chars ≈ $0.148 list / $0.170 buffered — measured across 6 real sessions, replacing the old 24,000-char maximum (which remains a safe worst-case bound, ~2.8× higher).
3. **Image generation is a single diagram per session** — one Gemini image at **$0.10 list / $0.115 buffered** (a conservative round-up from the ~$0.067 published list), a minor line that no longer scales with walkthrough depth.
4. **Everything carries a +15% buffer**, so the estimate stays deliberately conservative.
5. **Validate two things on real usage:** (a) the **average narrated characters** per session vs. the measured 4,758 / 12,611 / 8,685 figures — the dominant line; and (b) o3's reasoning-token count (Section 4.3).

Recommended checks before committing to a budget

- Confirm the **average narrated characters** per session (per route) against a larger sample than 6.
- Pull the **actual o3 output-token count** from a dozen real calculation sessions vs. ~21,000/session.
- Decide the Inworld **commitment tier** (Developer $17 → higher commit $12.50 → enterprise $10).

Sources

- OpenAI o3 API pricing ($2 / $8 per 1M tokens; hidden reasoning tokens billed at output rate): https://developers.openai.com/api/docs/pricing · https://pricepertoken.com/pricing-page/model/openai-o3
- Inworld TTS-2 pricing (Developer plan $17 / 1M characters; on-demand $25, tiered to $10 enterprise): https://inworld.ai/pricing · https://www.buildmvpfast.com/api-costs/ai-voice

- Measured TTS characters: `homework_tts_summary.pdf` / `homework_tts_char_count.py` (6 sessions, 2026-07-10).

Figures are grounded in the estimator's editable token / character / image assumptions; tune those to your real provider contracts and billed usage.